

# ZEYU (ALBAN) LI, Ph.D.

Materials Scientist | Polymers & Hydrogels | Ithaca, NY | 626-826-5327 | [zl788@cornell.edu](mailto:zl788@cornell.edu) | [zeyuli.net](http://zeyuli.net) | [linkedin.com/in/albanli](https://linkedin.com/in/albanli)

**Materials scientist** with full-cycle R&D experience in polymer formulation, hydrogel crosslinking, microparticle fabrication, and functional surface engineering. Research spans biodegradable polymeric microparticles, hydrogel composites for continuous biomanufacturing, tunable biopolymer hydrogel networks, and ultradurable superhydrophobic surfaces. Integrates computational and AI-assisted workflows into materials research. Six peer-reviewed publications, two patents.

## SKILLS

**Materials Fabrication & Processing:** 3D printing (DIW, SLA, FDM); Electrospinning; Thermal imprinting; Double-emulsion microencapsulation; Hydrogel formulation; UV photocrosslinking; Spin coating; Mold casting; Photolithography; Two-photon lithography; Etching; Nanopatterning; Freeze-drying; Roll-to-roll processing; Surface functionalization.

**Materials Characterization:** SEM; DLS; EDS; TGA; DMA; Optical microscopy; UV-Vis spectroscopy; Fluorescence spectroscopy; Fluorescence microscopy; Rheometry; Tensile testing; Contact-angle measurement.

**Programming & Data/Modeling:** Python; Data analysis & visualization (pandas, NumPy, seaborn); statistical DOE, Machine learning (TensorFlow, PyTorch); Java, ImageJ; Git/GitHub; AI tools (computational workflows, LLM-assisted data analysis, LLM-assisted research workflows, AI agent development).

**2D/3D Design & Visualization:** CAD; SketchUp; Autodesk 3ds Max; Blender; KeyShot; Adobe Illustrator, Photoshop, InDesign; Bootstrap Studio; Scientific figure production; Video/audio editing.

**Molecular & Bioengineering:** PCR; qPCR; IVT; Electrophoresis (PAGE, agarose); DNA/RNA extraction & purification; Cleanroom protocols; Chromatography; Enzymatic assays and workflows.

## EDUCATION

### Cornell University

Ithaca, NY

Biological and Environmental Engineering | Concentration: **Materials Science**, Biological Engineering, Bioenvironmental Engineering

Ph.D. | May 2026 | M.S. | Aug 2024 | M.Eng. | May 2020

### Hong Kong Baptist University

Kowloon, Hong Kong

Chemistry (Major) | Computer Science (Minor)

B.Sc. (Hons) | Nov 2019

## EXPERIENCE

### Graduate Research & Teaching Assistant

Feb 2021 – May 2026

DNA Materials Lab, Advisor: Prof. Dan Luo | Cornell University, Ithaca, NY

- **Formulated biodegradable PLGA polymeric microparticles for synthetic DNA encapsulation and environmental tracing;** produced via double-emulsion solvent evaporation with SEM-characterized particle morphology and size distribution. Scaled to gram-scale production and validated in a DoD-funded field deployment across 11 km<sup>2</sup>, confirming DNA detection 7 km downstream from sub-milligram release. (First-author paper, *ES&T*, 2025).
- **Engineered a PEG-based polymeric hydrogel composite on polyester mesh scaffold for continuous biomanufacturing;** UV-cured organogel with covalent DNA immobilization via NHS-amine bioconjugation. Systematically varied monomer concentration, crosslinker ratio, and functional loading, mapping non-monotonic process-structure-property relationships. Characterized morphology (SEM), network viscoelasticity (rheometry), spatial distribution (fluorescence microscopy); validated 100-day material stability, sustaining in vitro transcription of 6 RNA species from immobilized DNA templates. (Second-author paper, *Nature Sustainability*, in peer review.)
- **Developed biomass DNA biopolymer hydrogels with thermal, photochemical, and ionic crosslinking** from a low-cost feedstock; produced thermoreversible physical cross-linked hydrogels, photo-cleavable covalent cross-linked hydrogel, and instantaneously gelling ionic cross-linked hydrogels. Characterized gelation behavior, viscoelasticity, and fracture properties across 182 formulations by oscillatory rheometry, Instron tensile testing, and gravimetric swelling.

Demonstrated extrusion-based 3D printing and vascular self-healing in ceramic tile composites. (First-author paper, *ACADIA*, 2024).

- **Established perfluorocarbon as a high-capacity DNA/RNA capture medium** in a pod-based continuous-flow extraction platform; achieved 41.73 µg/µL adsorption capacity (800 times that of silica-based conventional methods), translating to ~209 mg DNA loading per pod with over 90% capture efficiency. Reduced projected consumables cost by ~91% compared with conventional ethanol-based method.
- **Co-instructed biomaterials classes and mentored 3 undergraduate researchers**; designed a 6-lecture computational tooling module integrating AI-assisted analysis into biomaterials workflows and organized a Biomaterials Hackathon. Teaching assistant across 7 engineering courses total. *Cornell CALS Outstanding Teaching Assistant Award*.

#### Senior Research Assistant & Undergraduate Research Assistant

Jun 2017 - Jan 2021

Microfabrication & Surface Materials Lab, Supervisor: Prof. Kangning Ren | Hong Kong Baptist University

- **Engineered monolithic perfluoropolymer superhydrophobic surfaces** via high-temperature soft imprinting and thermal fusion with master molds from photolithography, two-photon lithography, and etching; validated topology-specific superwettability and ultradurability across abrasion, compression, corrosive vapors, UV aging, thermal treatment, and 100,000+ folding cycles. (Third-author paper, *The Innovation* IF = 33.2; US Patent 11,839,998).
- **Designed and fabricated an automated weather-environment simulation chamber** integrating temperature and humidity control, supercooled droplet generation, adjustable droplet parameters, sample-stage cooling, and high-speed monitoring; incorporated an electrostatic surface-charging module to quantify voltage-dependent droplet behavior on superhydrophobic surfaces. Demonstrated ~25% reduction in droplet contact time under charged icing conditions. *Best Undergraduate Thesis Award*.

#### Research Exchange Trainee

Jun 2018 - Aug 2018

Molecular Basis of Disease Program | Georgia State University, Atlanta, GA

- **Characterized protein-DNA binding kinetics** using LSPR (localized surface plasmon resonance) assays. Measured how transcription factor PU.1 interacts with DNA. *Best Poster Presentation Award*.

#### PUBLICATIONS

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- Wang, D., **Li, Z.**, Li, J., Han, Y., Sun, T., Li, F., Liu, P. "A Chip Reactor for Perpetual Nucleic Acid Production and On-chip Information Processing." *Nature Sustainability*, in peer review (2026).
- **Li, Z.**, Ramón, C. L., Koeberle, A., et al. "Tracing Environmental DNA Transport in Large Lakes with Synthetic DNA Microparticles and Hydrodynamic Modeling" *Environmental Science & Technology* (IF = 12.4), (2025). DOI
- He, C., **Li, Z. (co-first authors)**, Wang, L. X., et al. "PolyTile 4.0: Self-healing Ceramic Tiles" *ACADIA* (2024). DOI
- Li, W., Chan, C. W., **Li, Z.**, et al. "All-perfluoropolymer, nonlinear stability-assisted monolithic surface combines topology-specific superwettability with ultradurability." *The Innovation* (IF = 33.2), 4(2), 100299, (2023). DOI
- **Li, Z.**, Liu, Z., Yang, X., Luo, D. "What the Records Don't Carry: A Position on Researcher-AI Co-Adaptation in Exploratory Laboratory Research." Under review at *ICML Workshop on AI for Science*, (2026).
- **Li, Z.** "Apprenticeship Without Proximity: An Agent Skills Architecture for Unwritten Experimental Judgment." Under review at *ACM CAIS Workshop on Agent Skills*, (2026).

#### PATENTS

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- Ren, K., Wu, H., Wang, Z., Yao, S., Ong, B., Li, W., **Li, Z.**, Sun, H., & Chan, C.W. "Crack engineering as a new route for the construction of arbitrary hierarchical architectures." US Patent 11,839,998, (2023).
- Li, Q., **Li, Z.**, & Lin, Z. "Reactor and method of spiral propulsion biomass continuous thermal cracking." Chinese Patent 201711214139.6 (2017).